INTERNATIONAL SEARCH REPORT

International application No.
PCT/JP2022/019726

A. CLASSIFICATION OF SUBJECT MATTER

G01M 99/00(2011.01)i
FI: G01M99/00

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01M99/00

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Published examined utility model applications of Japan 1922-1996
Published unexamined utility model applications of Japan 1971-2022
Registered utility model specifications of Japan 1996-2022
Published registered utility model applications of Japan 1994-2022

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	WO 2013/024571 A1 (PANASONIC CORPORATION) 21 February 2013 (2013-02-21) paragraphs [0034]-[0094]	1, 5-10
A		2-4
Y	JP 7-319507 A (BRIDGESTONE CORPORATION) 08 December 1995 (1995-12-08) paragraphs [0048]-[0061]	1, 5-10
A		2-4
A	JP 2017-101809 A (OKAYAMA UNIVERSITY) 08 June 2017 (2017-06-08) entire text, all drawings	1-10
A	US 2015/0369264 A1 (THE REGENTS OF THE UNIVERSITY OF MICHIGAN) 24 December 2015 (2015-12-24) entire text, all drawings	1-10
A	CN 104834808 A (QINGDAO UNIVERSITY OF SCIENCE & TECHNOLOGY) 12 August 2015 (2015-08-12) entire text, all drawings	1-10

☑ Further documents are listed in the continuation of Box C.

☑ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
15 July 2022	**02 August 2022**
Name and mailing address of the ISA/JP **Japan Patent Office (ISA/JP)** **3-4-3 Kasumigaseki, Chiyoda-ku, Tokyo 100-8915** **Japan**	Authorized officer Telephone No.

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C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 2011/086638 A1 (PANASONIC CORPORATION) 21 July 2011 (2011-07-21) entire text, all drawings	1-10
A	JP 4-081711 A (OLYMPUS OPTICAL COMPANY, LIMITED) 16 March 1992 (1992-03-16) entire text, all drawings	1-10

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/JP2022/019726

Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
WO 2013/024571 A1	21 February 2013	US 2013/0317647 A1 paragraphs [0099]-[0167]	
JP 7-319507 A	08 December 1995	(Family: none)	
JP 2017-101809 A	08 June 2017	(Family: none)	
US 2015/0369264 A1	24 December 2015	WO 2015/196057 A1 entire text, all drawings	
CN 104834808 A	12 August 2015	(Family: none)	
WO 2011/086638 A1	21 July 2011	US 2012/0133318 A1 entire text, all drawings CN 102781633 A	
JP 4-081711 A	16 March 1992	(Family: none)	